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EXACT POST-SELECTION INFERENCE, WITH APPLICATION TO THE LASSO

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We develop a general approach to valid inference after model selection. At the core of our framework is a result that characterizes the distribution of a post-selection estimator conditioned on the *selection event*. We specialize the approach to model selection by the lasso to form valid confidence intervals for the selected coefficients and test whether all relevant variables have been included in the model.

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Outline of our approach

The lasso and its selection event

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Introduction

Post-Selection

- High-Dimensional Data: In empirical studies, researchers start with a large pool of candidate variables, and they do not know a priori which are relevant. This is especially problematic when there are more variables than observations, since then the model cannot be identified.
- Post-Selection: In such settings, researchers often let the data decide which variables to include in the model.
  - Fit a linear model with all variables included, observe which ones are significant at level α, and then refit the linear model with only those variables included.
  - Information Criterion (AIC, BIC), Regularization (LASSO) etc.

Inference after Selection (1)

- However, a problem is arising with such post-selection: the p-values can no longer be trusted, since the variables that are selected will tend to be those that are significant.
- Intuitively, we are "overfitting" to a particular realization of the data.

Inference after Selection (2)

- Let us formalize the problem: Consider the standard linear regression setup, where the response $\boldsymbol{y} \in \mathbb{R}^n$ is generated from a multivariate normal distribution:

$$\boldsymbol{y} \sim \text{Normal}(\boldsymbol{\mu}, \sigma^2 \boldsymbol{I}_n), \tag{1}$$

  where

  - $\boldsymbol{\mu}$ is modeled as a linear function of predictors $\boldsymbol{x}_1, \cdots, \boldsymbol{x}_p \in \mathbb{R}^n$, and
  - σ^2 is assumed known.[1]

[1] We consider the more realistic case where σ^2 is unknown in Section 8.1.

Inference after Selection (3)

- We choose a subset $M \subset 1, \cdots, p$ and ask for the linear combination of the predictors in M that minimizes the expected error.
- That is,

$$\boldsymbol{\beta}^M \equiv \arg\min_{\boldsymbol{b}^M} \mathbb{E}||\boldsymbol{y} - X_M \boldsymbol{b}^M||^2 = X_M^+ \boldsymbol{\mu}, \tag{2}$$

  where $X_M^+ \equiv \left(X_M^T X_M\right)^{-1} X_M^T$ is the pseudo-inverse of X_M.
- Note that (2) implies that the targets β_j^M and $\beta_j^{M'}$ in different models ($M \neq M'$) are in general different.
- This is simply a restatement of the well-known fact: A regression coefficient describes the effect of a predictor, adjusting for the other predictors in the model.
- In general, the coefficient of a predictor cannot be compared across different models.

Inference after Selection (4)

- Thus, "inference after selection" is ambiguous in linear regression because the target of inference changes with the selected model.
  - Berk et al. (2013)
- In Section 2, we discuss several ways to resolve this ambiguity.

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REFERENCES

BENJAMINI, Y., HELLER, R. and YEKUTIELI, D. (2009). Selective inference in complex research. *Philos. Trans. R. Soc. Lond. Ser. A Math. Phys. Eng. Sci.* **367** 4255–4271. MR2546387

BENJAMINI, Y. and YEKUTIELI, D. (2005). False discovery rate-adjusted multiple confidence intervals for selected parameters. *J. Amer. Statist. Assoc.* **100** 71–93. MR2156820

BERK, R., BROWN, L., BUJA, A., ZHANG, K. and ZHAO, L. (2013). Valid post-selection inference. *Ann. Statist.* **41** 802–837. MR3099122

COX, D. R. (1975). A note on data-splitting for the evaluation of significance levels. *Biometrika* **62** 441–444. MR0378189

EFRON, B., HASTIE, T., JOHNSTONE, I. and TIBSHIRANI, R. (2004). Least angle regression. *Ann. Statist.* **32** 407–499. MR2060166

FISHER, R. (1956). On a test of significance in Pearson's Biometrika Tables (No. 11). *J. Roy. Statist. Soc. Ser. B.* **18** 56–60. MR0082773

FITHIAN, W., SUN, D. and TAYLOR, J. (2014). Optimal inference after model selection. Preprint. Available at arXiv:1410.2597.

JAVANMARD, A. and MONTANARI, A. (2013). Confidence intervals and hypothesis testing for high-dimensional regression. Preprint. Available at arXiv:1306.3171.

KNIGHT, K. and FU, W. (2000). Asymptotics for lasso-type estimators. *Ann. Statist.* **28** 1356–1378. MR1805787

LEEB, H. and PÖTSCHER, B. M. (2005). Model selection and inference: Facts and fiction. *Econometric Theory* **21** 21–59. MR2153856

References ii

LEEB, H. and PÖTSCHER, B. M. (2006). Can one estimate the conditional distribution of post-model-selection estimators? *Ann. Statist.* **34** 2554–2591. MR2291510

LEHMANN, E. L. and ROMANO, J. P. (2005). *Testing Statistical Hypotheses*, 3rd ed. Springer, New York. MR2135927

LOCKHART, R., TAYLOR, J., TIBSHIRANI, R. and TIBSHIRANI, R. (2014). A significance test for the lasso (with discussion). *Ann. Statist.* **42** 413–468. MR3210970

MILLER, A. (2002). *Subset Selection in Regression*, 2nd ed. Chapman & Hall/CRC, Boca Raton, FL. MR2001193

NEGAHBAN, S. N., RAVIKUMAR, P., WAINWRIGHT, M. J. and YU, B. (2012). A unified framework for high-dimensional analysis of M-estimators with decomposable regularizers. *Statist. Sci.* **27** 538–557. MR3025133

PÖTSCHER, B. M. (1991). Effects of model selection on inference. *Econometric Theory* **7** 163–185. MR1128410

PÖTSCHER, B. M. and SCHNEIDER, U. (2010). Confidence sets based on penalized maximum likelihood estimators in Gaussian regression. *Electron. J. Stat.* **4** 334–360. MR2645488

ROBINSON, G. K. (1979). Conditional properties of statistical procedures. *Ann. Statist.* **7** 742–755. MR0532239

SAMPSON, A. R. and SILL, M. W. (2005). Drop-the-losers design: Normal case. *Biom. J.* **47** 257–268. MR2145117

SILL, M. W. and SAMPSON, A. R. (2009). Drop-the-losers design: Binomial case. *Comput. Statist. Data Anal.* **53** 586–595. MR2654594

STOREY, J. D. (2003). The positive false discovery rate: A Bayesian interpretation and the q-value. *Ann. Statist.* **31** 2013–2035. MR2036398

References iii

TAYLOR, J., LOCKHART, R., TIBSHIRANI, R. J. and TIBSHIRANI, R. (2014). Post-selection adaptive inference for least angle regression and the lasso. Preprint. Available at arXiv:1401.3889.

TIAN, X. and TAYLOR, J. (2015). Asymptotics of selective inference. Preprint. Available at arXiv:1501.03588.

TIBSHIRANI, R. (1996). Regression shrinkage and selection via the lasso. *J. Roy. Statist. Soc. Ser. B* **58** 267–288. MR1379242

TIBSHIRANI, R. J. (2013). The lasso problem and uniqueness. *Electron. J. Stat.* **7** 1456–1490. MR3066375

TIBSHIRANI, R. J., RINALDO, A., TIBSHIRANI, R. and WASSERMAN, L. (2015). Uniform asymptotic inference and the bootstrap after model selection. Preprint. Available at arXiv:1506.06266.

VAN DE GEER, S., BÜHLMANN, P., RITOV, Y. and DEZEURE, R. (2013). On asymptotically optimal confidence regions and tests for high-dimensional models. Preprint. Available at arXiv:1303.0518.

WEINSTEIN, A., FITHIAN, W. and BENJAMINI, Y. (2013). Selection adjusted confidence intervals with more power to determine the sign. *J. Amer. Statist. Assoc.* **108** 165–176. MR3174610

ZHANG, C.-H. and ZHANG, S. S. (2014). Confidence intervals for low dimensional parameters in high dimensional linear models. *J. R. Stat. Soc. Ser. B. Stat. Methodol.* **76** 217–242. MR3153940

ZHONG, H. and PRENTICE, R. L. (2008). Bias-reduced estimators and confidence intervals for odds ratios in genome-wide association studies. *Biostatistics* **9** 621–634.

ZOLLNER, S. and PRITCHARD, J. K. (2007). Overcoming the winner's curse: Estimating penetrance parameters from case-control data. *Am. J. Hum. Genet.* **80** 605–615.

ZOU, H. and HASTIE, T. (2005). Regularization and variable selection via the elastic net. *J. R. Stat. Soc. Ser. B. Stat. Methodol.* **67** 301–320. MR2137327